

A Modified CFAR Algorithm Based on Object Proposals for Ship Target Detection in SAR Images

Hui Dai, Lan Du, Yan Wang, and Zhaocheng Wang

Abstract—Target detection for synthetic aperture radar (SAR) images has great influence on the successive discrimination based on the target regions. However, as a pixel-based method, the traditional constant false alarm rate (CFAR) detection could not work well for the ship target detection problem of multiple ship targets with different sizes in a SAR image, which is referred to as the multiscale situation. Moreover, it needs to use the clustering method on the pixel-level detection results to obtain the accurate target regions, which may merge two or more different targets into a target region. In this letter, a modified CFAR based on object proposals is proposed. We use the object proposal generator to generate a small set of object proposals with different sizes, and then use the proposal-based CFAR detector, where the extracted object proposals are regarded as the guard windows instead of setting fixed guard window, to detect the true positive object proposals. By introducing the object proposals as the variable guard windows in the CFAR detector, the proposed algorithm could gain good detection performance in the multiscale situation, since the missed detection resulting from the big differences between the sizes of the fixed guard window and ship targets can be avoided. Meanwhile, the proposed method can directly obtain the accurate target regions. The effectiveness of the proposed algorithm is verified using the measured SAR data.

Index Terms—Constant false alarm rate (CFAR) algorithm, object proposals, ship target detection, synthetic aperture radar (SAR).

I. INTRODUCTION

TARGET detection on synthetic aperture radar (SAR) images is a subtask of the SAR automatic target recognition (ATR) system, which typically consists of three stages: detection, discrimination, and classification or recognition [1]. As the first step of the whole SAR ATR system, target detection has great influence on the successive processing [2]. Since SAR images are widely used in ocean observation and ship monitoring, ship detection has been an important application of SAR images [3], [4]. Ship detection in SAR images has been studied extensively, and many ship target

detection methods have been developed [3]–[6], of which the two-parameter constant false alarm rate (CFAR) method is often used for ship target detection in SAR images [3], [4], [6].

The two-parameter CFAR proposed by the Lincoln Laboratory is actually a cell averaging CFAR technique based on the assumption of Gaussian background [2], [3]. There are mainly two disadvantages of the two-parameter CFAR algorithm for ship target detection in SAR images. First, the two-parameter CFAR needs to know the size of the detected ship first to determine the size of the guard window. It works well in the situation where a single target is present in locally homogeneous clutter [2]–[4]. If multiple ship targets with different sizes occur in a SAR image, which is referred to as the multiscale situation, the big differences between the sizes of the fixed guard window and ship targets will lead to inaccurate clutter distribution estimation due to the influence of interfering target pixels on the clutter region, which could result in missed detection. The usual way to solve the problem is to censor out interfering target pixels from the region of clutter estimation. Gao *et al.* [2] proposed a censoring approach based on the idea that the tailed part of the histogram of the SAR image represents target pixels. In this method, an empirical parameter, i.e., the censoring depth, has to be determined *a priori*. However, the setting of such a parameter requires knowledge about the size or the number of the targets in the image, and the optimal value is yet hard to obtain. Cui *et al.* [7] proposed an iterative censoring scheme (ICS) for target detection with SAR imagery, which iteratively updates the target maps for censoring. However, the convergence speed of the ICS is slow and the influence of target pixels on the sea clutter distribution estimation is not completely eliminated. To overcome the aforementioned two problems of the ICS, [8] proposed an improved iterative censoring detection scheme. From the experimental results in [2], [7], and [8], these methods work better than the traditional CFAR, but they usually detect more candidate target pixels, which may lead to more false alarms. Second, the CFAR algorithm uses a sliding window to detect the possible existing ship pixels. Then, the clustering method is implemented to obtain the accurate target regions, which is a difficult task for multiple ship targets located closely in a SAR image. In this situation, the pixels from two or more different targets may be remerged into a target region, which will degrade the detection performance and bring many difficulties to the following discrimination and recognition stages. So far, although there are many modified CFAR algorithms for ship target detection, they mainly focus on the former case and still need to use the clustering method to obtain the target regions.

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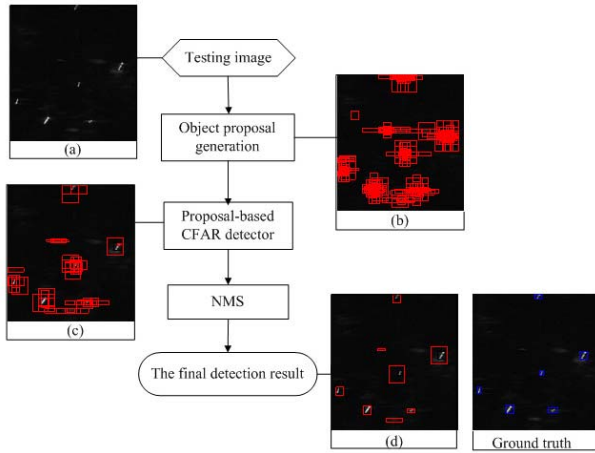


Fig. 1. Flowchart of the proposed ship target detection method.

According to the ATR system [1], the discrimination stage takes the target regions passed to it by the detection stage as its input. Therefore, the target regions-based evaluation on the detection results is of great significance in the detection stage of ATR system.

In recent years, object proposal generation has become a promising technique for many vision recognition tasks [9]–[11], especially for class-specific object detection in colorful optics images. Instead of handling with tremendous amount of bounding boxes in sliding windows fashion, object proposal generation can rapidly select much fewer candidate bounding boxes that cover most of the objects in the image. These candidate bounding boxes are called object proposals, which are with different sizes. Then, based on the object proposals, subsequent detector can be used to reduce the false positives (FPs) and obtain good detection result. Therefore, this technique benefits object detection from two aspects: speeding up the computation by reducing the bounding boxes and improving the detection accuracy by allowing the usage of more sophisticated learning machinery [12], [13].

To improve the detection performance of the two-parameter CFAR for ship target detection in the multiscale situation, the object proposals are introduced in SAR ship target detection, and a modified CFAR algorithm is proposed in this letter. We first use the object proposal generator to generate a small set of object proposals. Then, we use the proposal-based CFAR detector, in which we regard object proposals as guard windows rather than set a fixed guard window, to detect the true positive object proposals. Finally, the detection result is given by nonmaxima suppression (NMS) [14]. By introducing the object proposals as variable guard windows in the CFAR detector, the missed detection resulting from the big differences between the sizes of the fixed guard window and ship targets can be avoided. Thus, the proposed algorithm could improve detection performance in the multiscale situation. Meanwhile, we can directly obtain the accurate target regions to avoid the problem resulting from the clustering on the pixel-level detection results.

II. PROPOSED SHIP TARGET DETECTION METHOD

Given an image, we aim at detecting the ship targets. As shown in Fig. 1, the proposed approach consists of

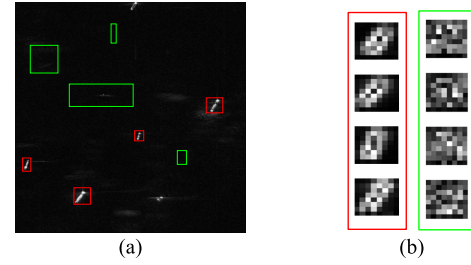


Fig. 2. Ship object (red) and nonobject (green) windows and their corresponding 64-D gradient features. (a) Source image. (b) 64-D gradient features.

three steps: 1) object proposal generation; 2) proposal-based CFAR detector; and 3) NMS processing [14]. An example on a test image is shown along each step for the better illustration purpose.

A. Object Proposal Generation

Object proposals are widely used in computer vision. Of the methods generating object proposals, one simple and efficient method is the binarized normed gradients algorithm [9]. This method stated that the generic objects with well-defined closed boundary can be discriminated by looking at the norm of gradients (NG), with a suitable resizing of their corresponding image windows into a small fixed size. Based on the computation reason, it resizes an image window to 8×8 and uses the NG as a simple 64-D feature to learn a generic objectness measure in a two stages cascaded support vector machine (SVM) framework.

For our SAR ship target detection problem, by resizing the ship target windows to different small fixed sizes, e.g., 16×16 , 8×8 , and 4×4 , we find that the 8×8 gradient features in intensity space become a good discriminate feature. The illustration of the ship object and nonobject windows and their corresponding 8×8 gradient features are demonstrated in Fig. 2. Although the ships have different sizes, they do share strong correlation in the gradient space. Thus, the 8×8 gradient features can be used to discriminate the object from nonobject, which can be utilized to predict the existence of objects effectively. The gradient features g are calculated by

$$g = \|g_x\| + \|g_y\| \quad (1)$$

where the horizontal gradients g_x and the vertical gradients g_y are calculated via the convolution of the 1-D masks $A = [-1, 0, 1]$ and A^T with image window in intensity space, respectively.

The flowchart of the object proposal generation is shown in Fig. 3. In the training stage, we first select the labeled ground truths in the training set and sample multiscale bounding boxes around each ground truth which have more than 0.5 intersection-over-union (IoU) overlaps with it as positive training samples. Background bounding boxes randomly sampled from each training image, which have less than 0.2 IoU overlaps with all the ground truths in the training image, are used as negative training samples. Then, 8×8 gradient features of the positive and negative training samples are used to learn an objectness measure mode $\mathbf{W} \in \mathbb{R}^{64}$ in a linear SVM [15]. In the detection stage, we first resize the test image to different quantized and predefined sizes and calculate the gradient map of each resized image. Then, an 8×8 sliding window is used on

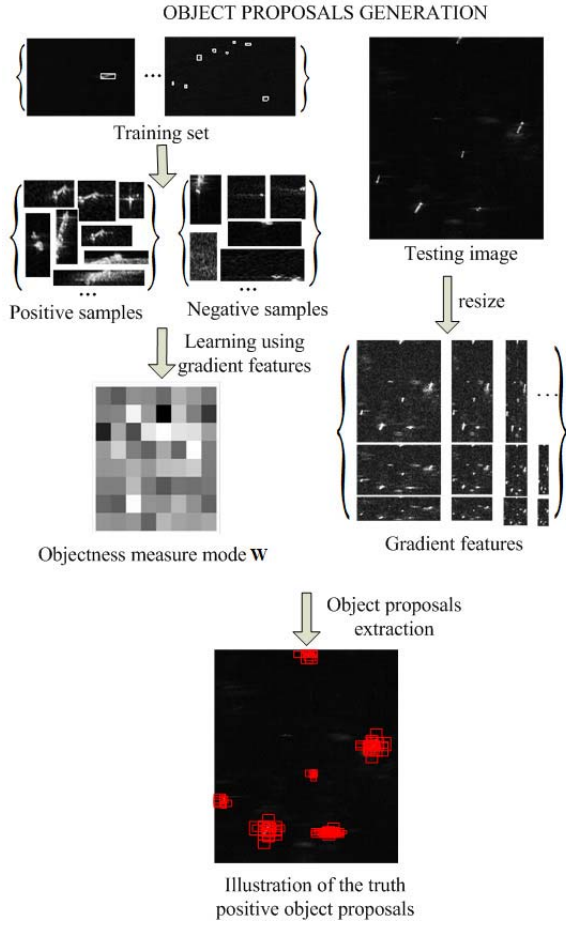


Fig. 3. Framework of the proposed object proposal generation.

the each gradient map to obtain the 64-D gradient features g_l , where the index $l = (i, x, y)$ with i and (x, y) denoting the size index of the resized test image and the position of the sliding window in the gradient map, respectively. The gradient features g_l are scored with the learned mode $\mathbf{W}$

$$s_l = \langle \mathbf{W}, g_l \rangle \quad (2)$$

where s_l and $\langle \cdot \rangle$ represent the filter score of g_l and the inner product operator, respectively. Finally, for each size i , we select a small set of proposals using NMS [14], which will be further introduced in Section II-C.

B. Proposal-Based CFAR Detector

Object proposal generation is only the preprocessing of our detector. It ensures little false negatives at the cost of numerous FPs. The number of object proposals is determined on the condition that the object proposal generation can achieve 100% recall rate on the training set when these object proposals are used. In the subsequent detection, we need to remove the FPs and improve the detection precision. Different to the traditional two-parameter CFAR, our proposed CFAR detector is based on object proposals and regards the object proposals as the guard windows to detect the true positive object proposals. The variable guard windows instead of the fixed guard window can improve the detection performance in the multiscale situation. The proposal-based CFAR detector is shown in Fig. 4, where the red dotted boxes show the main

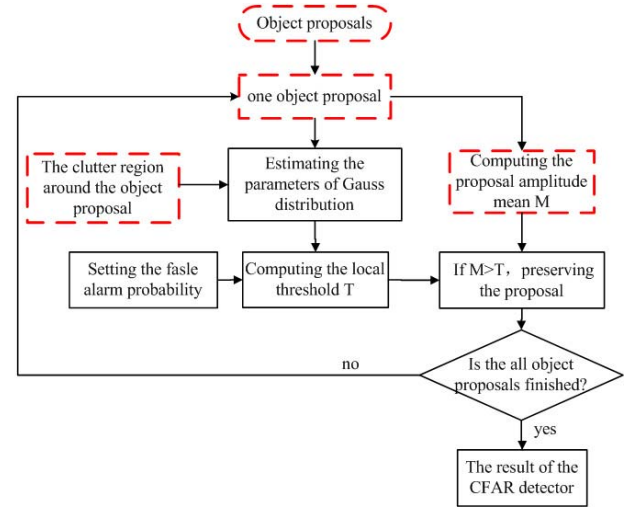


Fig. 4. Flowchart of the proposal-based CFAR detector.

differences between our detector and the traditional CFAR detector.

An object proposal whose amplitude mean M stands out from the surroundings is preserved as the true positive object proposal according to

$$M_{\text{clutter}}^{\text{target}} \geq T \quad (3)$$

$$T = \hat{\mu}_c + \hat{\sigma}_c \cdot K_{\text{CFAR}} \quad (4)$$

$$K_{\text{CFAR}} = \Phi^{-1}(1 - p_{\text{fa}}) \quad (5)$$

where $\hat{\mu}_c$ and $\hat{\sigma}_c$ are the estimated mean and the standard deviation of the local clutter amplitude, respectively, T is the local threshold adaptively obtained for object proposal, K_{CFAR} is a constant related to set false alarm probability p_{fa} , and $\Phi^{-1}(\cdot)$ is the inverse of the standard normal cumulative distribution function.

C. NMS Processing

Using the proposal-based CFAR detector described earlier, we get multiple overlapping object proposals for a ship target as shown in Fig. 1(c). Then, we employ NMS [14] to further eliminate the overlapping proposals around each target. We sort the proposals by their amplitude mean M , and greedily select the highest ones while skipping proposals that are at least 10% covered by previously selected proposals. The final detection result of the example is shown in Fig. 1(d).

III. EXPERIMENTS

A. Experimental Images

In this section, we demonstrate the performance of our proposed ship target detection method on the three measured SAR images shown in Fig. 5¹, where the sizes are 833×3687 , 3931×3872 , and 2100×1750 , respectively. In our experiment, we cut the first two large images into 28 small images. Then, we select eight small images in which the sizes of ship targets vary greatly for training. The other small images and the third

¹Fig. 5(a) and (b) is from RADARSAT-2 data, which were purchased from the Center for Earth Observation and Digital Earth, Chinese Academy of Sciences, and Fig. 5(c) is our measured data, which was provided by our cooperative institution.

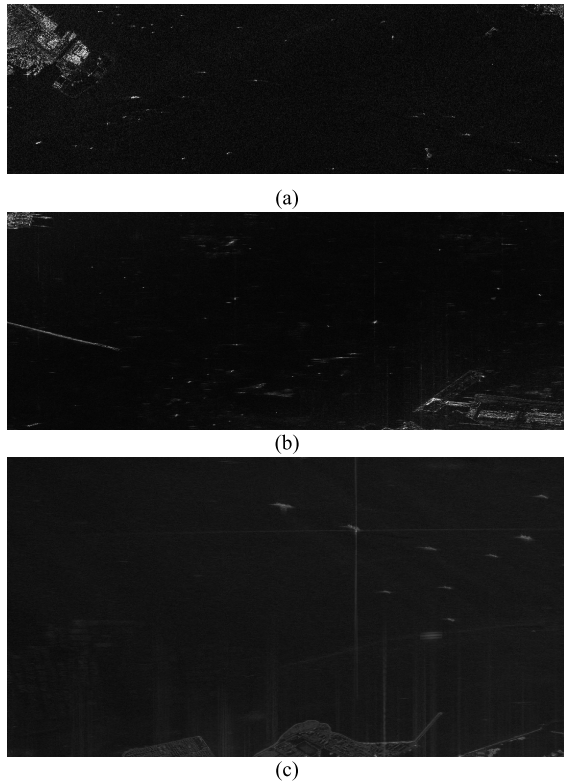


Fig. 5. Source images. (a) and (b) are from RADARSAT-2 data, (c) is our measured data.

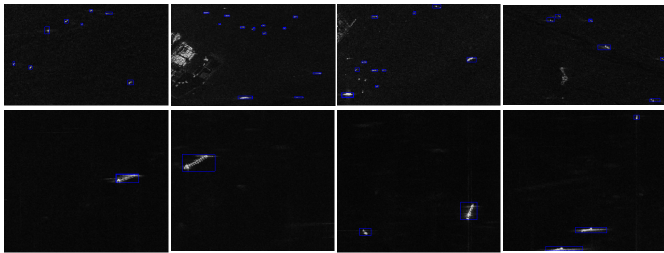


Fig. 6. Training set with ship targets manually marked. The sizes of images in the first row are 333×1000 and 500×500 in the second row.

large image are used for testing. The training set is shown in Fig. 6, where the blue rectangles are the ground truths manually marked by visual inspection.

B. Detection Results

According to the ATR system, the detected target regions in the detection stage are inputted into the SAR target discrimination step. In [2], the target regions-based detection results are given. Referring to the ATR system and [2], the detection results and the evaluation based on target regions are given in this letter.

We take the traditional two-parameter CFAR and the improved ICS [8] as the reference methods in this letter to compare the detection performance with our proposed method. To compare the reference methods fairly, both the two-parameter CFAR and the improved ICS are under the Gaussian distribution assumption and use the same clustering method in [2] to obtain the target regions in the following experiment.

We set $p_{fa} = 10^{-6}$ in our proposed method and the reference methods. Due to the length limitation, we only demonstrate the detection results on six test images shown

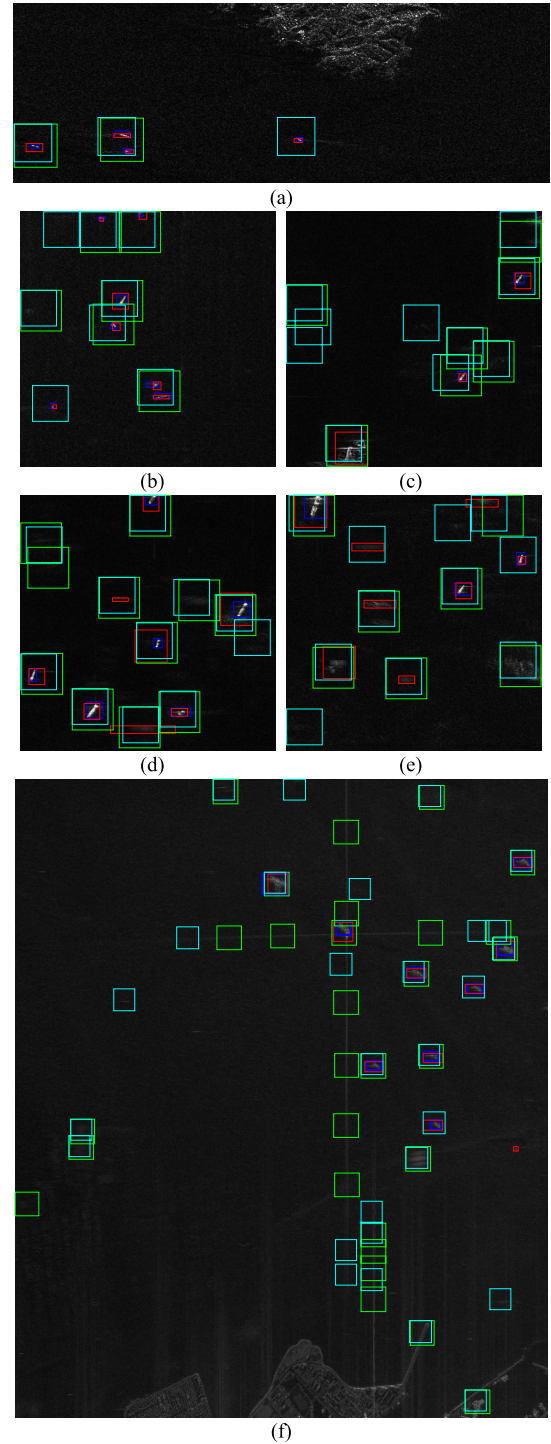


Fig. 7. Ship target detection results on the test images. The sizes of (a) and (f) are 333×1000 and 1390×1750 , respectively. The sizes of (b)–(e) are 500×500 . The blue rectangles are ground truths. The red rectangles, green rectangles, and cyan rectangles are extracted by our proposed method, the tradition CFAR, and the improved ICS, respectively.

in Fig. 7. The blue rectangles are the ground truths. The red rectangles, green rectangles, and cyan rectangles are the detection results of our proposed method, the two-parameter CFAR, and the improved ICS, respectively. We can see that our proposed method has no false negatives, i.e., missing targets, and every detection window has only one target. However, the reference methods have false negatives in some test images, e.g., Fig. 7(a), (b), (e), and (f). In addition, one detected target

TABLE I
DETECTION PERFORMANCE COMPARISON BETWEEN OUR METHOD
AND THE REFERENCE METHODS ON TEST IMAGES IN FIG. 7

Test images in Fig. 7		(a)	(b)	(c)	(d)	(e)	(f)
Ground truth: Number of ship targets (NP)		4	7	2	6	3	9
Our proposed method	Number of detected ship targets	4	7	2	6	3	9
	Number of true positive windows (TP)	4	7	2	6	3	9
	Number of false positive windows (FP)	0	0	1	2	5	1
Two-parameter CFAR	Number of detected ship targets	3	6	2	6	2	7
	Number of true positive windows (TP)	2	5	2	6	2	7
	Number of false positive windows (FP)	0	1	5	5	5	22
Improved ICS	Number of detected ship targets	4	7	2	6	3	8
	Number of true positive windows (TP)	3	6	2	6	3	8
	Number of false positive windows (FP)	0	2	7	5	8	20

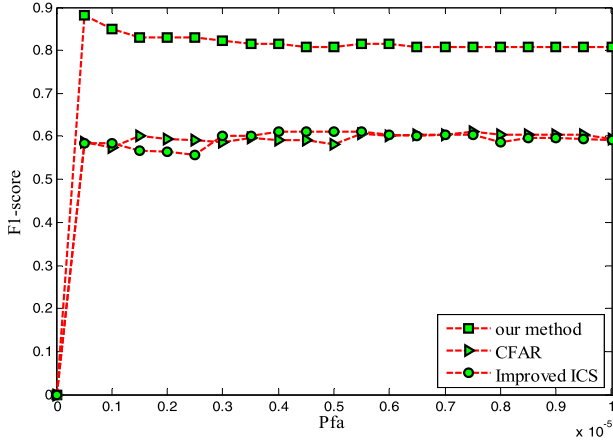


Fig. 8. Average F1-score curves of our method and the reference methods on test images in Fig. 7 under different false alarm probabilities.

window may contain two or more targets when they locate closely via the reference methods, e.g., Fig. 7(a) and (b).

Table I gives the quantitative comparison of the detection results between our method and the reference methods on test images in Fig. 7. Our method is based on object proposals and any objects undetected at this stage cannot be recovered later. From Table I, we can see that the number of true positive windows is equal to the number of ground truths (NP) for each test image via our method, which demonstrates that there are no false negatives, i.e., missing targets. Therefore, the object proposal generation in our method achieves 100% recall rate via 1650 proposals on each test image. In addition, we can also see that the number of FPs, i.e., the number of false alarm targets, via our method are less than those via the two reference methods for the test images in Fig. 7(b)–(d) and (f).

In addition, we evaluate the detection performance via F1-score. The F1-score is defined as

$$\text{F1-score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (6)$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad \text{recall} = \frac{\text{TP}}{\text{NP}} \quad (7)$$

where TP, FP, and NP are defined in Table I. Fig. 8 shows the average F1-score curves of our method and the reference methods on test images in Fig. 7 under different false alarm probabilities. From Fig. 8, we can see that our method can get higher F1-scores than the reference methods in the range of $p_{fa} = [0, 10^{-5}]$, which indicates that our method has the higher detection accuracy than the reference methods.

IV. CONCLUSION

In this letter, we propose a modified CFAR algorithm based on object proposals for ship target detection in SAR images. First, we use an object proposal generator based on gradient features to generate a small set of object proposals. Then, a proposal-based CFAR detector is used to detect the true positive object proposals. Finally, the final detection result is obtained via NMS. Experimental results on the measured SAR images demonstrate that our proposed method could get the better detection performance than the pixel-based detection methods in the multiscale situation for ship target detection.

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